Hygiene Investment Urgency Index

A data-driven WASH investment priority tool

for policymakers at the World Bank, UNICEF, and WHO

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BUS32120 · Final Project

The Decision Problem

Where should WASH funding go?

87 countries

× **4** WASH interventions

= 348 funding choices

Today's reality

Allocation decisions are often made without integrated, cross-source WASH evidence.

Funding may not flow to the countries — or the interventions — where the health return is greatest.

Our goal: replace guesswork with an evidence-based priority ranking.

Our Approach

Five stages — from raw data to a ranked priority list

1. **Data Sources**
   DHS · World Bank
   OECD · WHO/UNICEF
2. **SQL Merge**
   LEFT JOIN on
   iso3 + year
3. **EDA**
   Quality checks
   & patterns
4. **Regression**
   Quantify each
   WASH lever
5. **Urgency Index**
   Country-level
   priority ranking

My part covers steps 1–2 today; my teammates will walk you through 3–5.

Data Sources & SQL Pipeline

Five complementary sources, merged into one country-year table

Sources

DHS Program API
Household WASH & child health outcomes

World Bank (core)
GDP per capita, life expectancy, population

World Bank (literacy)
Female literacy rate

World Bank (health)
Health expenditure %, urbanization

OECD
Official development assistance (ODA)

Merge

```
LEFT JOIN on iso3 + year
```

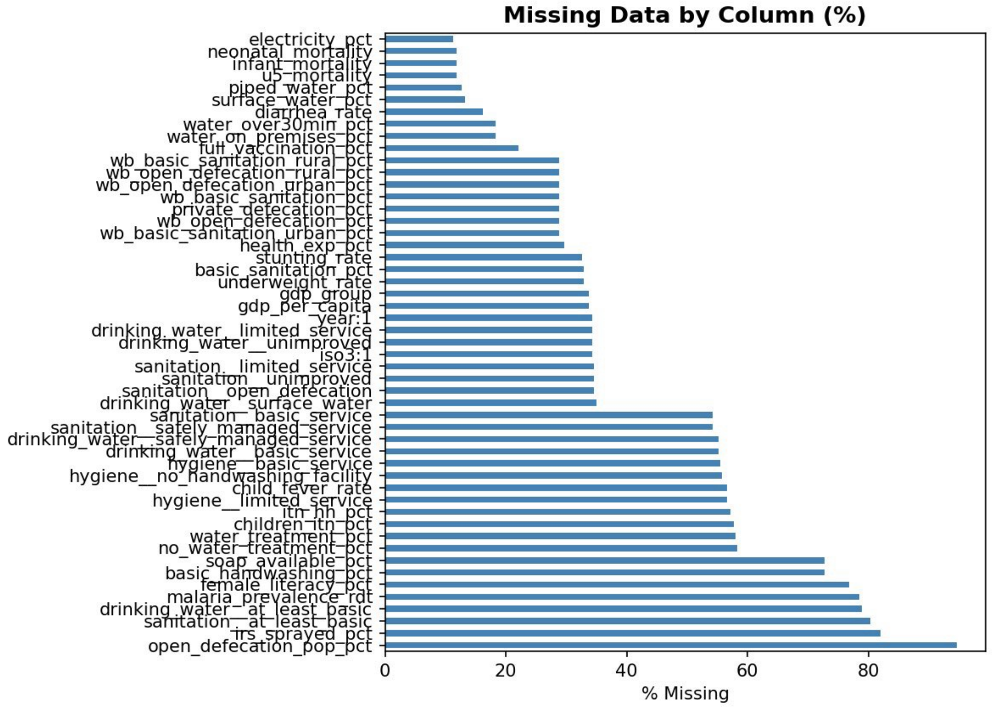
Each row = one country in one survey year. Joins preserve all DHS rows; macro columns fill in where available.

659	90	40
country-year observations	countries	years (1985–2024)

Why SQL: reproducible joins, transparent logic, easy to audit by anyone reviewing the analysis.

Data Quality: What You Should Know

Honest caveats before we get into the analysis



COVERAGE

Uneven by country

DHS rounds vary — some countries appear 27 times, others only once.

MISSINGNESS

Structural, not random

Gaps reflect which DHS modules were fielded — not measurement error.

EXTREMES

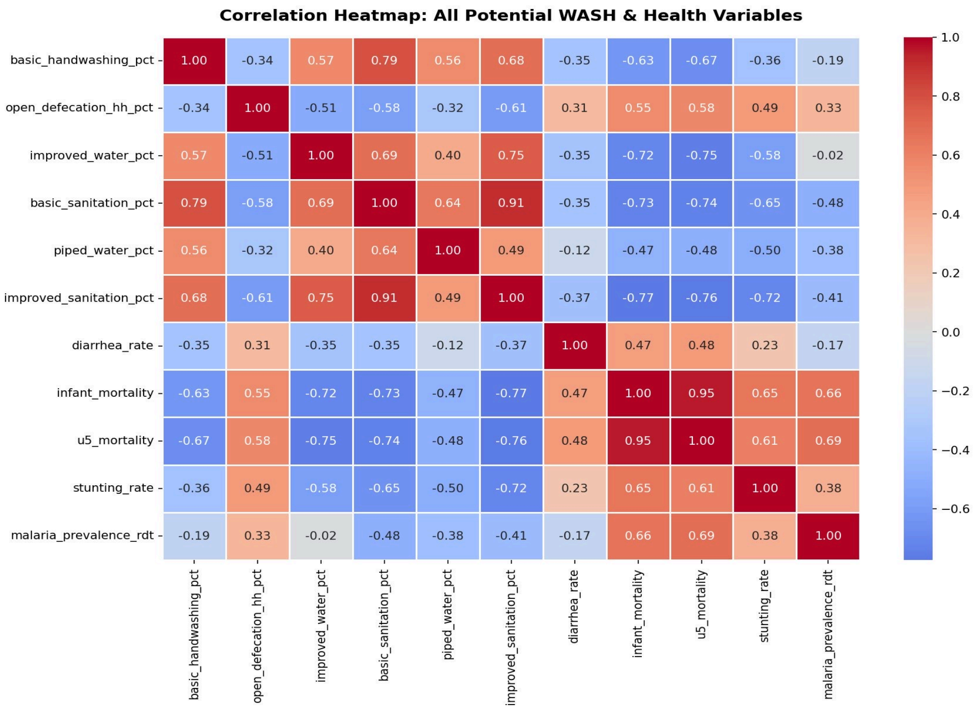
Real, not anomalies

Niger 1992: u5 mortality = 322 per 1,000 — kept in (real conditions).

Implication: trust regional patterns more than country-specific values where DHS coverage is thin.

What the Data Tells Us

Correlation heatmap: WASH indicators vs. health outcomes



Open-defecation **r = 0.58**

with under-5 mortality

basic sanitation **r = −0.76**

with under-5 mortality

improved water **r = −0.75**

with under-5 mortality

Basic handwashing **r= -0.67**

with under-5 mortality

Modeling Approach

Quantify how much each WASH dimension actually moves child mortality

Models

Linear regression × 3

Predicts each continuous outcome:

- diarrhea rate
- infant mortality
- **under-5 mortality ← weights for the Index**

Logistic regression × 1

Binary: high-risk if infant mortality > median.

Robustness check — 81.2% accuracy.

Key Setup Choices

309

country-year observations
after regional-mean imputation

Why standardize features?

Variables operate on different scales (GDP in thousands; WASH % bounded 0–100). Standardizing to mean=0, std=1 makes coefficients directly comparable — so we can use them as weights in the Urgency Index.

Regression Results

Standardized coefficients — which WASH lever moves mortality most?

WASH Variable	Diarrhea rate	Infant mortality	Under-5 mortality ★
Improved water %	−1.06	−10.31	**−21.93**
Open defecation (HH) %	+0.96	+3.60	**+7.75**
Basic handwashing %	−1.74	+0.21	**−5.60**
Basic sanitation %	+0.27	−3.10	**−2.52**
Model R^2	*0.159*	*0.583*	***0.634***

★ U-5 model has the highest R^2 and is used as weights for the Urgency Index.

KEY INSIGHT

Improved water

|−21.9|

is roughly

3× larger

than the next biggest WASH lever

Each unit of standardized water access change is associated with ~22 fewer deaths per 1,000 live births.

Building the Urgency Index

Combining health impact with a country's room for improvement

$$\mathbf{Urgency_i} = \Sigma \left(\mathbf{Gap_i} \times |\boldsymbol{\beta_i}| \right)$$

GAP

Room for improvement

Distance from full coverage in each WASH dimension, normalized 0–1.

Example: a country at 20% water access has a much larger gap than one at 90%.

COEFFICIENT |β|

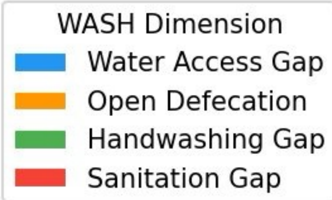
Health impact per unit

From the under-5 mortality regression ($R^2 = 0.63$ — best-fitting model).

Tells us how much child mortality moves per unit change in that dimension.

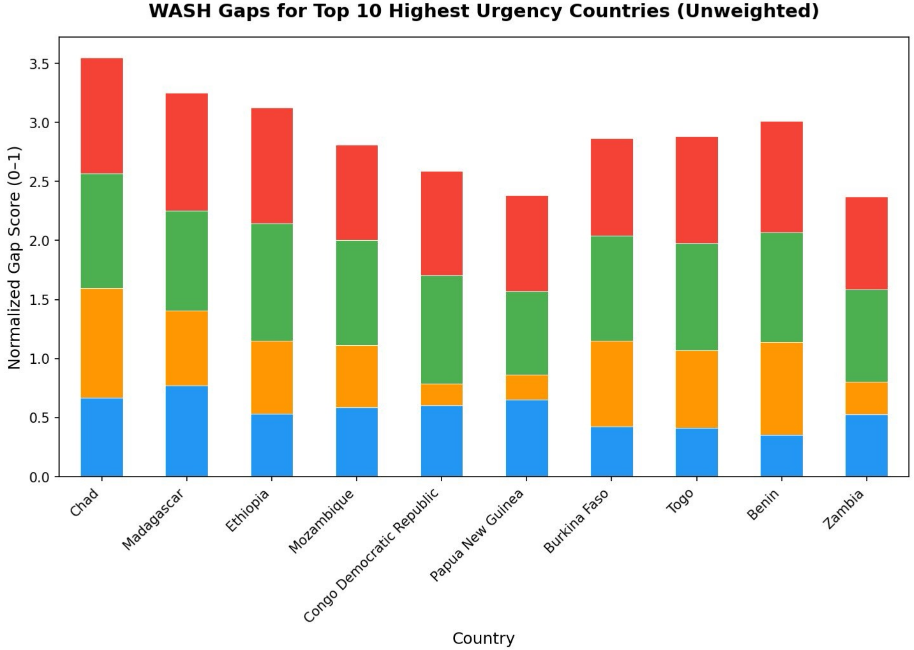
Final index rescaled to 0–100 — where 100 = highest investment urgency.

Where to Invest, Within Each Country



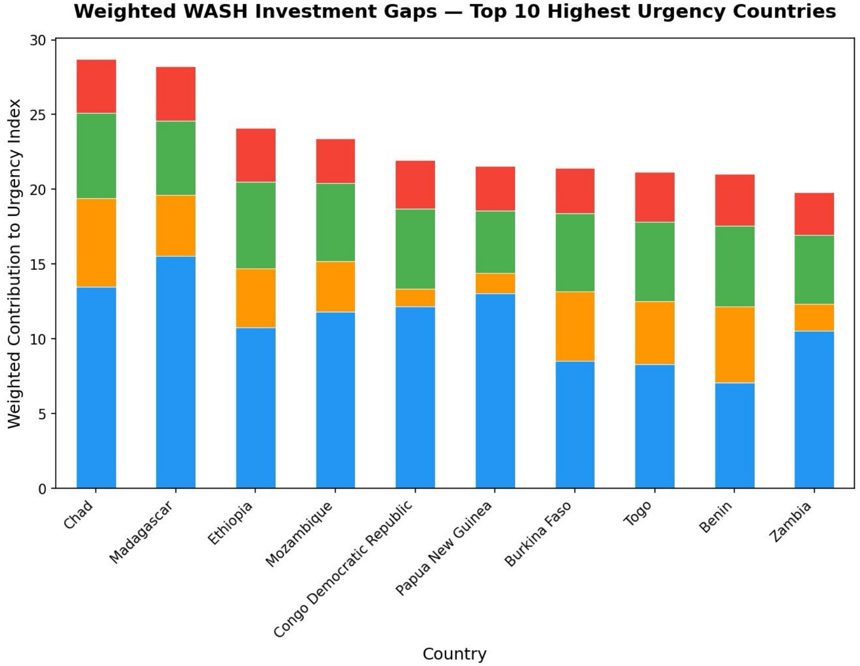
Adding the regression weights changes which dimension matters most

RAW GAP — by size only



Sanitation gap looks dominant

WEIGHTED — by health impact

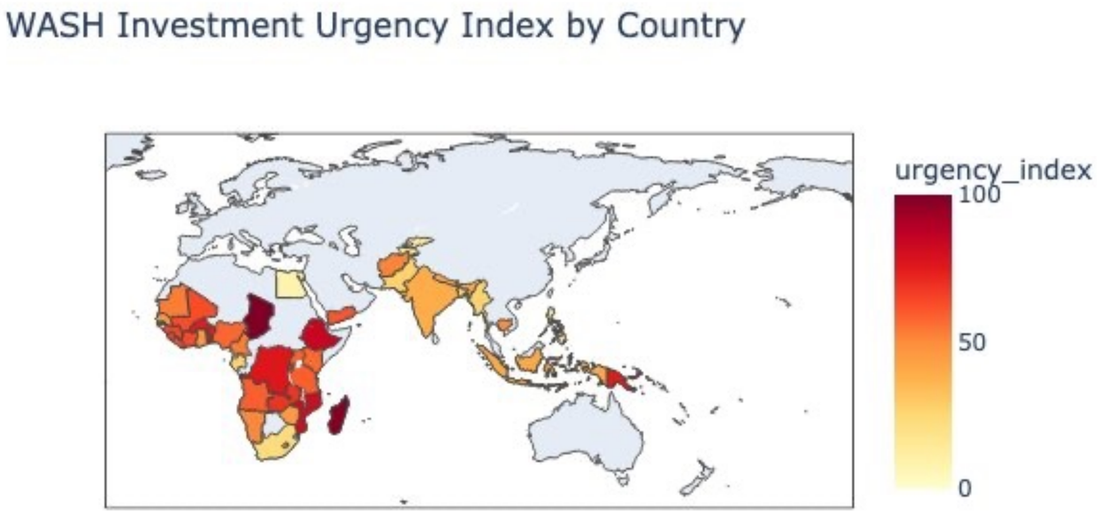


Water dominates — biggest mortality lever

Implication: a country's biggest raw gap isn't always its highest-return intervention. Weighting reveals that water comes first.

Where the Need Concentrates

Urgency Index by country — Sub-Saharan Africa dominates



TOP 10 HIGHEST URGENCY

1	Chad	SSA	100.0
2	Madagascar	SSA	98.4
3	Ethiopia	SSA	83.8
4	Mozambique	SSA	81.3
5	Congo DR	SSA	76.2
6	Papua New Guinea	OCE	74.8
7	Burkina Faso	SSA	74.4
8	Togo	SSA	73.4
9	Benin	SSA	73.0
10	Zambia	SSA	68.7

9 of the top 10 are in Sub-Saharan Africa; Papua New Guinea is the lone outlier.

Recommendations for WB · UNICEF · WHO

Three concrete priorities from the index

1 GEOGRAPHIC PRIORITY

Concentrate on SSA + PNG

9 of 10 highest-urgency countries are in Sub-Saharan Africa. Oceania has one hotspot — Papua New Guinea.

2 INTERVENTION PRIORITY

Water first, within every country

Improved water has the largest mortality coefficient (~3× the next lever). Highest return per dollar.

3 INDEX IS VALIDATED

Higher urgency = higher real deaths

High-tier countries (n=17) average 129 deaths/1,000 — vs. 53 in Low. The index tracks reality.

Validation: average actual under-5 mortality by index tier

129	HIGH (n=17)	91	MEDIUM (n=23)	53	LOW (n=13)

Questions?